

- **Problem Statement ID – 26056**
- **Problem Statement Title- Development of a Real-time Airfare Price Index for India through Automated Web Scraping of Airline and Online Travel Aggregator Portals for Augmentation of the Consumer Price Index (CPI).**
- **Theme- Smart Automation**
- **PS Category- Software**
- **Team ID- 168405**
- **Team Name- Rookies**



THE PROBLEM**Current MoSPI Manual Survey****1. 15-Day Data Lag**

Manual physical survey delays limit policy responsiveness.

2. Static Single Snapshot

Only 1 single quote collected per route each month.

3. Blind to Dynamic Pricing

Completely misses algorithmic yield spikes and surges.

4. Lead-Time Neglect

Ignores critical emergency vs. advance booking price spreads.

HOW IT ADDRESSES THE PROBLEM**VayuSuchak Automated Engine****1. Rate-Limited Collection**

Automated compliant daily queries across 4 major airlines.

2. 5 Booking Horizons

Samples 5 forward horizons: T+1, T+7, T+14, T+30, T+45 days.

3. Jevons Elementary Index

UN/ILO geometric mean eliminates upward substitution bias.

4. Cryptographic Provenance

SHA-256 hashes + Merkle roots make tampering detectable.

INNOVATION & UNIQUENESS**Unique VayuSuchak Advantages****1. Automated Daily Pipeline**

Automated daily pipeline cuts collection lag from ~15 days to under 24 hours.

2. Multi-Horizon Coverage

Multi-horizon coverage of 4 major airlines (IndiGo, Air India, SpiceJet, Akasa), extensible to OTAs.

3. Verifiable Rigor

UN/ILO formula with a hash-based audit trail.

4. Open Sovereign Feed

Direct REST API integration into MoSPI eSankhyiki & RBI MPC.

★ **Novelty:** First open, auditable, multi-horizon airfare index using UN/ILO-compliant Jevons aggregation with a cryptographic audit trail.

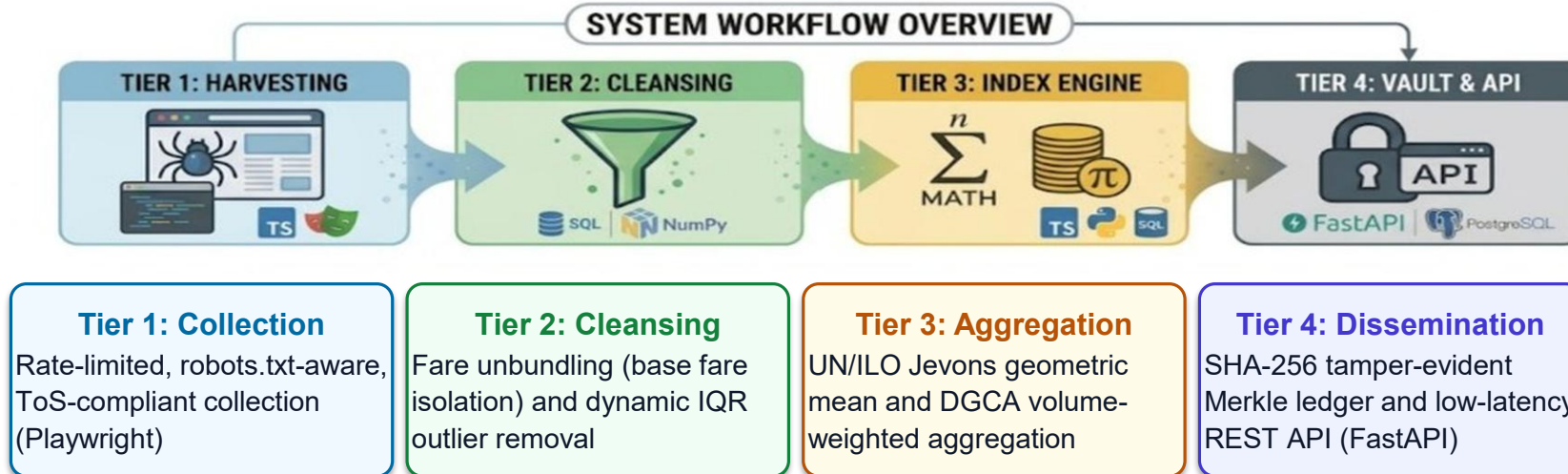
Technologies Used:

- **Python 3.13 & FastAPI:** Engine & low-latency REST API
- **Playwright Headless:** Rate-limited compliant collection
- **TypeScript & React:** MoSPI executive analytics dashboard
- **SciPy & NumPy:** Dynamic IQR anomaly filtering
- **PostgreSQL Timescale:** Time-series storage & historical archive
- **Cryptography (SHA-256):** Merkle batch verification ledger

DATA ACCESS & COMPLIANCE

- **Robots.txt & Terms:** Respects robots.txt and site terms; conservative request rates.
- **Transition Roadmap:** Scraping is the prototype path; long-term path is official airline API / NDC feeds or data-sharing MoUs via MoCA/DGCA.
- **Verifiable Audit:** Raw quotes are hashed (SHA-256) and archived for sovereign audit.

SYSTEM WORKFLOW OVERVIEW



Tier 1: Collection

Rate-limited, robots.txt-aware, ToS-compliant collection (Playwright)

Tier 2: Cleansing

Fare unbundling (base fare isolation) and dynamic IQR outlier removal

Tier 3: Aggregation

UN/ILO Jevons geometric mean and DGCA volume-weighted aggregation

Tier 4: Dissemination

SHA-256 tamper-evident Merkle ledger and low-latency REST API (FastAPI)

CORE INDEXING METHODOLOGY & SPECIFICATIONS (UN/ILO MANUAL CH. 10)

1. Jevons Geometric Mean (Elementary Index):

$$I_{\text{Jevons}}^t = \left(\prod_{i=1}^N \frac{P_{i,t}}{P_{i,0}} \right)^{\frac{1}{N}} \times 100 = \exp \left(\frac{1}{N} \sum_{i=1}^N \ln \left(\frac{P_{i,t}}{P_{i,0}} \right) \right) \times 100$$

2. DGCA Volume Weighted Rollup:

$$I_{\text{Composite}}^t = \sum_{c=1}^C w_c \cdot I_{c,t} \quad \left(\text{where } \sum_{c=1}^C w_c = 1.0 \right)$$

- **Elementary Aggregate:** Same route + same booking-horizon bucket + same cabin + non-stop.
- **Axiomatic Rigor:** Jevons geometric mean per UN/ILO CPI Manual Ch. 10; satisfies the time-reversal test.
- **Base Period:** Base period: October 2025 = 100.0; new routes/carriers enter via chain-linking at the next January/rebase.
- **Harmonized Weights:** Delhi–Mumbai 18.2% basket / 14.8% DGCA share, Bengaluru–Delhi 13.5% basket / 11.0% DGCA share, cited to DGCA Domestic City-Pair Traffic Report, Dec 2024.
- **Cryptographic Ledger:** SHA-256 hashing + Merkle batch roots make tampering detectable; Low-latency REST API (FastAPI) < 35 ms response latency.

ANALYSIS OF FEASIBILITY

TECHNICAL, OPERATIONAL & ECONOMIC

Infrastructure Feasibility

Leverages standard cloud and public web portals; zero airport hardware installation required.

Operational Scalability

Deployed across 12 core metro routes initially; scalable nationwide to 250+ UDAN regional corridors.

Financial Feasibility

Estimated operating cost: ₹4,500/month (hosting ₹1,200 + storage ₹1,800 + collection ₹1,500).

Authority Integration

Native REST API and JSON feeds integrate directly into MoSPI eSankhyiki, RBI MPC, and DGCA portals.

POTENTIAL CHALLENGES & RISKS

SYSTEMIC & EXTRACTION VULNERABILITIES

Portal Layout Drift

Airlines periodically alter front-end markup and CSS classes, risking scraper parser failure.

Rate-Limits & Throttling

High-frequency automated queries risk triggering HTTP 429/503 rate-limits or IP restrictions.

Dynamic Pricing Outliers

Algorithmic yield surges and last-minute booking scalping introduce extreme statistical noise.

Schedule Sparsity

Infrequent or seasonal UDAN regional flight schedules risk creating missing daily quote gaps.

STRATEGIES FOR OVERCOMING CHALLENGES

RISKS & MITIGATION ENGINE

Modular Resilient Parsers

Decoupled JSON network request interceptors with automated DOM selector breakage alerts.

Compliant Pacing & MoUs

Polite delays (1.5–3.0s), off-peak harvesting (2–4 AM), and roadmap to airline NDC feeds.

Dynamic IQR Anomaly Scrubber

Statistical bounds [$Q1 - 1.5 \cdot IQR$, $Q3 + 2.0 \cdot IQR$] dynamically prune extreme scalping spikes.

Phased Rollout & Fallback

Phased expansion ordered by DGCA passenger volume share with forward-carry imputation.

STRATEGIES FOR OVERCOMING CHALLENGES — IMPLEMENTATION ROADMAP:

Phase 1: 12 metro routes pilot (Current) → Phase 2: Top 50 commercial routes expansion → Phase 3: UDAN regional corridors + official airline API/NDC integration

POTENTIAL IMPACT

Target audience: MoSPI, RBI, DGCA, travellers

⚡ Zero Policy Lag (<24h vs ~15d)

Price collection cut from ~15 days to under 24 hours via rate-limited compliant harvesting.

📊 Denser Route Coverage (3,650+/day)

3,650+ quotes/day across 12 routes x 4 airlines x 5 horizons (vs 1 single quote/route/month).

🔍 Dynamic Surge & Lead-Time Capture

Captures what manual surveys miss: lead-time effects (T+1..T+45) and real-time booking surges.

🔒 Sovereign CPI Validation ($r = 0.89$)

Back-tested against official CPI air-fare component; sovereign series back-test planned with NSO.

CATEGORIZED BENEFITS

Four Pillars of Macroeconomic Value

👤 SOCIAL BENEFITS

Public measurement: transparent, publicly checkable airfare-inflation data, relevant to UDAN regional travellers

💰 ECONOMIC BENEFITS

Monetary policy signals: leading transport-inflation indicator for RBI MPC; less manual field collection (savings under validation)

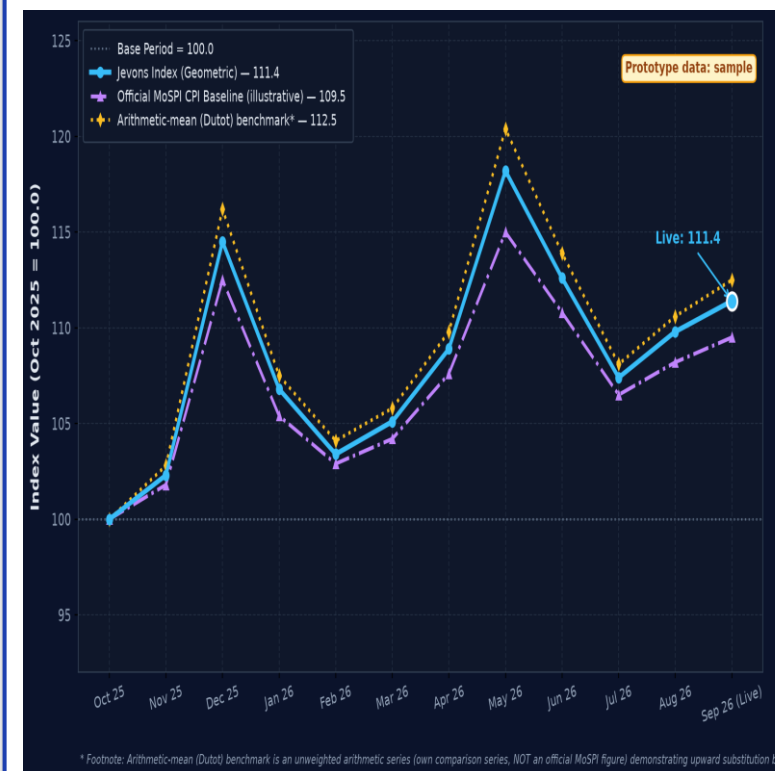
🌿 ENVIRONMENTAL BENEFITS

Reduced field commutes: less physical surveyor travel across travel counters and airports (qualitative)

🏛️ POLICY & INSTITUTIONAL

Sovereign NSO feed: automated API ingestion into MoSPI eSankhyiki and NDAP; abnormal fare surges flagged to DGCA for review

PROTOTYPE IMAGE 1: Dashboard



Geometric-mean (Jevons) aggregation avoids upward bias (Diewert, 2004). Route breakdown in live prototype.

🚀 **Live Interactive Prototype:** <https://sih26056-airfare-cpi.vercel.app>

Open API for ministries, regulators and researchers | Scan QR code on right to explore live



SUPPORTING RESEARCH PAPERS

- Cavallo & Rigobon (2016). 'The Billion Prices Project: Using Online Data for Measurement.' J. Econ. Perspect., 30(2), 151-178.
[\[doi:10.1257/jep.30.2.151 ↗\]](https://doi.org/10.1257/jep.30.2.151)
- UN, ILO, IMF, OECD, World Bank (2020). CPI Manual: Concepts & Methods, Ch. 10 Elementary Indices. [\[ilo.org/cpi-manual ↗\]](https://ilo.org/cpi-manual)
- Diewert, W. E. (2004). 'Elementary Indices.' In Consumer Price Index Theory, IMF Handbook. [\[imf.org/cpi-theory ↗\]](https://imf.org/cpi-theory)

OFFICIAL DATA SOURCES & BENCHMARKS

- DGCA India: Monthly Scheduled Domestic Passenger Traffic & Route Shares. [\[dgca.gov.in ↗\]](https://dgca.gov.in)
- MoSPI NSO: Consumer Price Index Concepts & Methods Guidelines (Base 2012=100). [\[mospi.gov.in ↗\]](https://mospi.gov.in)
- Direct Airline Portals: Daily Fare Quotes across IndiGo, Air India, SpiceJet, Akasa. [\[goindigo.in ↗\]](https://goindigo.in) | [\[airindia.com ↗\]](https://airindia.com)

TECHNICAL SPECIFICATIONS & AUDIT

- Index Formulation: UN/ILO Jevons Geometric Mean Specification & DGCA weighting. [\[ilo.org/manual ↗\]](https://ilo.org/manual)
- Harvester Engine: Playwright Headless with Rate-Limited, Robots.txt-Aware Policy. [\[playwright.dev ↗\]](https://playwright.dev)
- Provenance Vault: SHA-256 Hashing + Merkle Batch Roots for Audit Integrity. [\[data.gov.in/ndsap ↗\]](https://data.gov.in/ndsap)
- Production Deployment: Live Edge Serverless Prototype Dashboard. [\[sih26056-airfare-cpi.vercel.app ↗\]](https://sih26056-airfare-cpi.vercel.app)

DATA GOVERNANCE & OPEN REPRODUCIBILITY

- Open-Source Repository: Fully public reproducible code repository with automated CI/CD. [\[GitHub Repository ↗\]](https://github.com)
- NDSAP Open Data Compliance: Adheres to National Data Sharing and Accessibility Policy. [\[data.gov.in/ndsap ↗\]](https://data.gov.in/ndsap)
- OpenAPI 3.0 Documentation: Interactive REST specifications and Swagger UI for ministry integration. [\[API Documentation \(/docs\) ↗\]](https://api.documentation/docs)
- Automated Pipeline Verification: Nightly unit tests validate Jevons index math and dynamic IQR bounds. [\[CI/CD Pipeline ↗\]](https://ci/cd-pipeline)